

Yuezhexuan(Jason) Zhu

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Education

University of California, San Diego

Sep 2025 – Jun 2027 (Expected)

Master of Science in Computer Science | GPA: 3.90 / 4.00

University of Toronto

Sep 2020 - Jun 2025

Bachelor of Science in Computer Science | GPA: 3.96 / 4.00

Highlights of Qualifications

Languages: C/C++, Python, Golang, Java, TypeScript & JavaScript, Shell Script, Lua

Frameworks & Tools: Git, Docker, Postman, SpringBoot, PyTorch, React, Angular, GitHub Actions, Svelte

Cloud & Databases: MySQL, PostgreSQL, Redis, RocketMQ, Kubernetes, Elasticsearch, Nginx

Internship

Backend Developer Intern, ByteDance, Shanghai

Apr 2025 – Sep 2025

- Architected and deployed a standardized 3D Secure (3DS) encryption & validation mechanism for the Global Payment Network, safeguarding over **100k transactions daily** against fraud while ensuring regulatory compliance
- Identified and patched a critical financial risk in the refund logic; collaborated with QA and Operations to implement a robust fix that automated the accurate tracking of **100+ daily channel refunds**, directly mitigating potential revenue loss
- Spearheaded the channel standardization for Worldpay, refactoring legacy channel-specific logic into unified pipelines and configurations to reduce technical debt and improve system maintainability
- Collaborated with Product and QA teams to optimize multiple payment channels, enhancing system robustness and significantly reducing transaction failure rates

Software Engineer Intern, IBM Canada

May 2023 – Aug 2024

- Engineered core features for IBM Watson Code Assistant for Z ([WCA4Z](#)), developing a standardized automation tool to modernize legacy mainframe COBOL code into maintainable Java, accelerating digital transformation for enterprise clients
- Designed and implemented an automated end-to-end CI testing pipelines using Cypress and Jenkins, eliminating repetitive manual testing and saving over **30 engineering hours per release cycle**
- Delivered full-stack enhancements for the “Analyze-Refactor” module using React and Springboot, ensuring seamless integration with existing CI/CD infrastructure

Lab Intern, Pardee Lab at University of Toronto

May 2022 - Sep 2022

- Established a high-performance GPU server on cloud for [AlphaFold2](#) and budget-efficient server infrastructure
- Constructed machine learning guided bio-similar protein prediction [pipeline](#) with a high level of automation

Project & Research

Main Developer & Maintainer, [DiagWiki](#)

Dec 2025 – Present

- Developed diagram-centered, Agentic AI tool that autonomously analyze codebases to generate interactive Mermaid architecture diagrams, facilitating easier documentation and system understanding
- Engineered a high-performance Local RAG (Retrieval-Augmented Generation) system using Ollama and vector stores to enable context-aware querying of project documentation without requiring external API keys or cloud dependencies
- Built a decoupled architecture with a FastAPI backend for heavy AI orchestration and a Svelte frontend for responsive UI

Main Developer & Maintainer, [UTLeetcode](#)

Dec 2022 - Present

- Designed and maintained a full-stack web platform adopted by 40+ University of Toronto students; tracked registered users' coding progress & statistics using Leetcode's API and managed data using database for hourly granularity + user ranking
- Automated backend updates with GitHub Actions and deployed via [GitHub Pages](#) for high extensibility and availability

Interactive Visualization Engineering

Apr 2020 - Present

- Representative published works: [Flow Field](#), Fourier Transform [interactive platform](#), and [Deep Dream Transformer](#)
- Showcased creative applications of software engineering to a broad audience via [personal website](#) and YouTube [channel](#)